

Python Data Visualization Assignment Report

Itziar Morales Rodríguez

April 12, 2026

Contents

1	Introduction	2
1.1	Methodology	2
1.2	Sources	2
1.3	AI Usage	2
2	Dataset	2
2.1	Analysis	3
2.2	Processing	3
2.3	Research Questions	4
2.3.1	Question 1: Activity Composition	4
2.3.2	Question 2: Geographic Patterns	4
2.3.3	Question 3: Weekly Activity Assessment	4
3	Storytelling	4
4	Visualization Summary	4
5	Library Choices & Best Practices	4
6	Conclusions	5

1 Introduction

The aim of this project is to explore data visualization using Python, from selecting the data all the way to telling a story through it's visualization.

1.1 Methodology

The project itself followed the following process to achieve the previously mentioned goal:

1. Dataset selection
2. Data analysis and processing
3. Questions derived from the dataset
4. Visualization types needed to answer each question and storytell
5. Further data processing needed for each visualization
6. Library selection for each visualization
7. Package visualizations within a combined story narrative that tells a story about the data

1.2 Sources

Numerous sources are used through out the project such as:

- AI
- Documentation
- Forums
- Literature
- Lecture notebooks

Regardless of the source, these are indicated through bibliography/references or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage

A point was made to use AI in this project in a manner in which it's answers are scrutinized and reviewed, to amke sure hallucinations are not taken into account and answers are not just implemented without a second thought. This way, an activate usage of AI is incentivized with certain restraints in place.

AI being used for this project was also hosted locally, using smaller models to do so to encourage reproducibility (since prompts are also included within the notebook and report).

2 Dataset

Data for this project was required to not have been selected by a different student. To minimize the risk of this happening, activity from a Gamrmin Forerunner 245 was chosen, reducing the chances of another student having the same dataset. The data located in `./data/activities.csv` consists of:

- **Shape:** 31 columns & 180 rows
- **Date span:** 21 February 2021 - 30 March 2026 (5 years)
- **Data types:** Dates, numerics, categorical, geospatial (Title contains locations)

- **Key columns:** Date, Activity Type, Time, Distance, Avg HR
- **Activity types:** Running, Walking, Cycling, Strength, Swimming, etc.

Having 31 columns allows for many different questions to be asked about the dataset, specially since some of these columns allow for other data fields to be extracted (activity intensity from Avg HR). The limited number of rows (180) whilst still being a decent amount, allows it's content to be analyzed in a more hands on way, being able to be more critical about each specific row and actively engage in processing in a more granular fashion. The dates spanning 5 years back allows for a story to be told through time, and analyze factors such as consistency. Lastly, varying data types allows for distinct types of visualizations that require specific units to be used such as coordinates (from location).

2.1 Analysis

Before asking questions about the dataset, it was further analyzed using both the `df.info()` function to check how the data was imported into the pandas dataframe. A small AI analysis was invoked using qwen3.5:9b for the prompt which was the following:

Read the linked CSV file and analyze it, create a table with the following fields:

- N: Column number
- Column: Column name
- Description: Content description (what the data in the column usually) looks like), excluding null indicators
- Null indicator: Indicator that seems to be used for when the data is null
- Data type: Suggested data type to be used when reading CSV in Pandas.

The table generated was then used as a starting point although it's content was scrutinized and interpreted correctly, assessing whether or not the AI had hallucinated when generating it. Some important conclusions reached with the help of this analysis (done in section 2.1 of `data_viz_notebook.ipynb`) where:

- Some columns weren't really needed or used, such as *Favorite*
- "-" was used as a null indicator on most columns
- Correct data types for each column
- Some columns belonged to specific activity types, such as *Avg. Swolf* for swimming

2.2 Processing

Key processing steps (section 3 of `data_viz_notebook.ipynb`):

- Replaced '-' nulls with NaN
- Converted Time/Distance to numeric (regex parsing)
- Extracted locations from Title
- Created WHO-equivalent activity minutes¹ from Avg HR thresholds
- Weekly aggregation for time series analysis
- Activity Type simplification (treadmill running, trail running, running are all running)

Libraries: Pandas,² NumPy,³ re⁴ (regex)

Some of the data processing was effectuated in order to satisfy the data requirements of some of the visualizations, which will be explained in the following section.

2.3 Research Questions

Three questions were formulated based on data exploration:

1. What types of activities dominate the dataset and their frequencies?
2. How do activity patterns vary by location?
3. Do recorded activities show case "enough" weekly exercise?

2.3.1 Question 1: Activity Composition

Question 1 is a good starting point because it allows for the superficial exploratory analysis of the data. This facilitates the understanding of what activity types end up shaping most of the data, and is a good starting point that is taken into account when further exploring other questions.

2.3.2 Question 2: Geographic Patterns

Question 2 proceeds naturally from question 1, as it further dives into understanding the nature of the data, in this case, classifying it into geographic location rather than categorical activity types. Furthermore, it also enables the use of different visualization methods due to the unique geographic data type, location.

2.3.3 Question 3: Weekly Activity Assessment

Question 3 is purposefully subjective to some degree, to allow for a variety of approaches to visualizing and analysing the data, alongside more liberty when it comes to choosing what narrative to pursue in storytelling. This allows for many different charts to be used, and for data to be used in creative ways.

3 Storytelling

The focus in Storytelling will be the following: A fitness conscious user who has a garmin watch, has measured their activities multiple years. With questions 1 and 2 we can come to understand whether the user participates in varied activity types regardless of their location (suggesting that they might sustain their fitness throughout holidays). Through the third question, we can come to understand whether this data being recorded, suggests that the user is engaging in enough fitness activity to be considered healthy, what periods has this happened in, and is the data trending in a positive direction.

We get to explore the fitness journey of an individual through visualizations, leaving us with a deep understanding of what their situation is, what is good about their current fitness, and how they've improved through out time.

4 Visualization Summary

5 Library Choices & Best Practices

Matplotlib: Core plotting, manual layouts, custom colormaps. Used for precise control (heatmaps, timelines).

Seaborn: Statistical plots (barplot, regplot). Simplified complex visualizations while maintaining customization.

ipywidgets: Interactivity for year comparison. Enhanced exploratory analysis.

Best practices applied (Unit 1):

- Color encoding follows perceptual principles (sequential for magnitude)
- Minimal ink: removed grids, tight spacing
- Clear hierarchy: prominent thresholds, trend lines
- Accessibility: high contrast, clear typography

Question	Charts	Chart Type	Library
Q1	Fig.	Horizontal bar	Seaborn Matplotlib
Q2	Fig.	Map	Plotly Bokeh
Q3	Fig.	Heatmap	Seaborn
		Timeline	Matplotlib
	Fig.	Lollipop (interactive)	ipywidgets Matplotlib
	Fig.	Scatter+trend	Seaborn Matplotlib

Table 1: Visualization summary meeting all requirements (3+ questions, 5+ charts, 3 libraries, 1+ data source).

6 Conclusions

The analysis reveals:

- Running dominates training (47%) but consistency improved 2025–2026
- WHO targets met sporadically; 2026 shows upward trend but needs sustained 300+ min weeks
- Location patterns suggest trail runs yield higher intensity

Future work: integrate weather data via API, add interactive dashboard with Plotly Dash.

References

- [1] World Health Organization. *WHO guidelines on physical activity and sedentary behaviour*. Accessed: 2026-04-11. 2020. URL: <https://www.who.int/publications/i/item/9789240015128>.
- [2] *Pandas*. Version 3.0.1. 2026. URL: <https://pandas.pydata.org/>.
- [3] NumPy Steering Council. *Python Interactive Widgets*. Version 2.4.3. 2026. URL: <https://numpy.org/>.
- [4] *re*. 3.14.3. 2026. URL: <https://docs.python.org/3/library/re.html>.